

TOOLS AND TECHNOLOGIES USED

DSA STRING & FLOW ALGORITHM STUDIO

Console program (Java) and frontend website (HTML, CSS, JavaScript)

Project Title	Dictionary-Based Error Detection and Suggestions in Text Processing System	
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JAVA	WEB	DATA	TOOLS
Console program	HTML • CSS • JS	Dictionary • Corpus	VS Code • Live Server

1. Programming Languages

A. Console Program Java

- **Java (JDK 11+)** — all algorithms, the menu-driven console program, file handling and evaluation.

B. Frontend Website HTML • CSS • JavaScript

- **HTML5** — page structure, navigation tabs, input areas and result panels.
- **CSS3** — dark theme, grid layouts and responsive styling.
- **JavaScript (vanilla)** — Trie, edit distance, KMP, Aho-Corasick, Max Flow and all UI logic.

2. Software / IDEs

A. Console Program Java

- **Visual Studio Code** — used to write the Java code.
- **VS Code Console (integrated terminal)** — used to compile (javac) and run (java) the program.

B. Frontend Website HTML • CSS • JavaScript

- **Visual Studio Code** — used to write the HTML, CSS and JavaScript files.
- **Live Server (VS Code extension)** — runs index.html in the browser with automatic reload.

3. Frameworks / Libraries

A. Console Program Java

- **No external libraries** — only the Java standard library: java.util (collections, Scanner, Random), java.io / java.nio (file reading), java.util.regex (tokenising) and java.math.BigInteger.
- **Algorithms written from scratch** — Trie, Damerau-Levenshtein, Soundex, KMP, Aho-Corasick, Max Flow (Edmonds-Karp, Dinic).

B. Frontend Website HTML • CSS • JavaScript

- **No frameworks or libraries** — no React, Angular, Vue or jQuery; only vanilla JavaScript and browser DOM APIs.
- **Inline SVG** — draws the Max Flow bipartite graph.
- **performance.now()** — measures algorithm timings in the live benchmark.
- **Google Fonts** — Inter and JetBrains Mono, loaded via link tag.

4. Databases / Platforms

A. Console Program Java

- **No database** — data is read from local text files in the corpus/ folder.
- **dictionary.txt** — 370,105 words; main vocabulary.
- **common_words.txt** — about 1,070 common words used for ranking suggestions.
- **bigrams.txt** — 457 common word pairs for limited context correction.
- **test_data.csv** — 70 labelled pairs used by the Evaluator.
- **input_docs/** — sample text files for file-based spell checking.
- **Platform** — runs on the local JVM.

B. Frontend Website HTML • CSS • JavaScript

- **No database or server** — all data is embedded in the page as JavaScript arrays.
- **dictionary_data.js** — 370,105-word dictionary and the common-words list.
- **Platform** — runs in the web browser's JavaScript engine.

5. Hardware / IoT Components

A. Console Program Java

- **Not applicable** — software-only project; runs on a laptop or desktop with JDK installed.

B. Frontend Website HTML • CSS • JavaScript

- **Not applicable** — software-only project; runs on any computer with a modern web browser.

6. Other Tools / Platforms

A. Console Program Java

- **README.md** — run steps, class table, complexity notes and viva answers.
- **.vscode (launch.json, settings.json)** — VS Code run configuration and Java source/output paths.
- **run.command** — macOS script that compiles and runs the program.

B. Frontend Website HTML • CSS • JavaScript

- **README.md** — run steps, file breakdown and viva answers.
- **.vscode (launch.json, tasks.json)** — browser launch and local server tasks.
- **run.command** — macOS script that starts a Python local server and opens the site.

PROJECT TECHNOLOGY SUMMARY

Console program vs frontend website

	Console Program	Frontend Website
Language	Java (JDK 11+)	HTML, CSS, JavaScript
Editor	Visual Studio Code	Visual Studio Code
How it runs	VS Code console: javac, then java	Live Server in the browser
Libraries	None (Java standard library only)	None (vanilla JS, inline SVG)
Data source	corpus/ text files	Embedded dictionary_data.js
Database	None	None
Hardware / IoT	Not applicable	Not applicable

Both versions use no external libraries, so every DSA algorithm is implemented in the project itself.